

Class 33 Cardinality of Sets

Goals

1. Know how to decide if two sets have the same number of elements.
2. Know the difference between countable and uncountable sets.
3. Prove the rational numbers are countable but the reals are uncountable.
4. Know Cantor's Theorem

Terminology: The cardinality of a set, A , is

Definition: Two sets A and B have the same cardinality if

Question: Is it possible for a set and a proper subset to have the same cardinality?

Definition: A set, A , is countable if

Examples of countable sets:

Any subset of a countable set is countable.

Any set that contains an uncountable set is uncountable.

The set of all points in the xy-plane with positive integer coordinates is a countable set.

$(1,4)$	$(2,4)$	$(3,4)$	$(4,4)$
$(1,3)$	$(2,3)$	$(3,3)$	$(4,3)$
$(1,2)$	$(2,2)$	$(3,2)$	$(4,2)$
$(1,1)$	$(2,1)$	$(3,1)$	$(4,1)$

Define a bijection
between the elements
in $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$
by creating a list
of all the points

Let Q_0 be the set of rational numbers in the interval $[0,1)$.
Let Q_1 be rationals in $[1,2)$ and so forth for $Q_2, Q_3, \dots$

Create a list of elements in Q_0 to prove that the set Q_0 of rational numbers in $[0,1)$ is countable.

Since we are able to construct a list of elements in Q_0 we know there is a bijection f from $\mathbb{N}$ to Q_0

